

HIGH VISIBILITY

REFLECTIVE MATERIALS

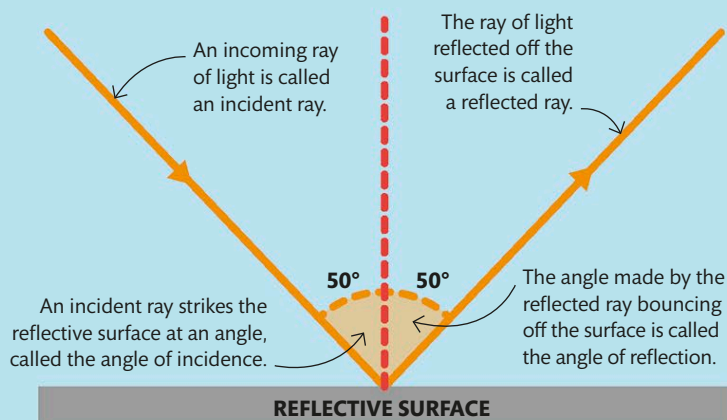
High-viz, or high-visibility, clothing is essential for cyclists and people who work in safety-critical jobs such as construction. It allows them to be easily seen when it is dark or if there is a lot of activity going on around them. High-viz clothing is made using materials that are retroreflective (reflect light directly back to its source), making the object much easier to see. These materials are also used to make road signs easy to see even in the dark.

Easy to spot

The retroreflective strips on this child's bicycle and clothing shine brightly in an oncoming car's headlights, making them easy to spot on a dark winter evening.

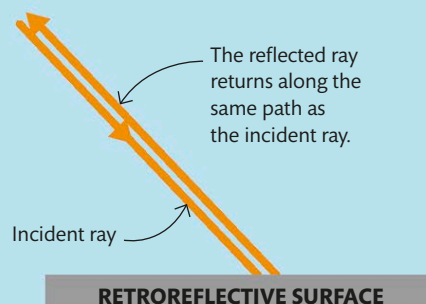
REFLECTION

Most objects scatter light in all directions when it hits them, but when light bounces off shiny or reflective surfaces, it bounces back in a predictable way. The reflected light makes the same angle as the incoming, or incident, light.



Retroreflection

Retroreflective surfaces are covered with tiny glass beads or prisms, which reflect light back in the same direction it comes from. Retroreflective materials cause light to be precisely directed back to its source so the object appears to shine brightly.



The eyes of a cat appear to shine in the dark because of a retroreflective layer in its eyes.